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New Laryngoscopic Technique by use of Fiber Optics

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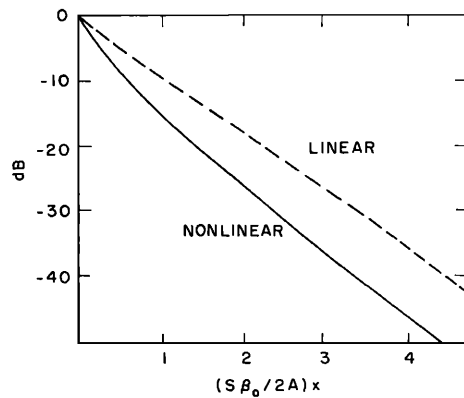


FIG. 1. Example of pressure distribution in a duct with nonlinear wall impedance.

the beginning of the duct, the static pressure P , and $\gamma = c_p/c_v$. Using this expression in the integral in Eq. 1, we obtain the following relation between the sound pressure p and the position x along the duct.

$$\frac{p}{p_0} \left(\frac{1+\delta}{1+\delta(p/p_0)^2} \right)^2 = \exp \left(\frac{-S\beta_0 x}{2A} \right). \quad (4)$$

For sufficiently large values, $x \gg (2A/S\beta_0)$, we get

$$p \approx p_0(1+\delta)^{-2} \exp(-S\beta_0 x/2A) \quad x \gg (2A/S\beta_0),$$

which shows that, as a result of the nonlinearity, the pressure is reduced by a factor of $(1+\delta)^{-2}$ corresponding to a level reduction of $40 \log(1+\delta)$. In Fig. 1, we have plotted $20 \log(p/p_0)$ as a function of $(S\beta_0/2A)x$ for $\delta=1$, which clearly shows the nonlinear change of level with distance. For a given pressure p_0 at the beginning of the duct, the nonlinear part of the level reduction increases with decreasing θ_0 ; but it should be realized that the transition from nonlinear to linear behavior occurs at an x value given approximately by $(2A/S\beta_0)$, which also increases with decreasing θ_0 . Also, it should be emphasized that the present analysis is restricted to the frequency region below the resonance frequency of the duct lining.

The general (three-dimensional) analysis of sound propagation in ducts with nonlinear boundary conditions is considerably more complex than the transmission-line approximation, even if we restrict ourselves to simple types of duct excitation, and we do not pursue this problem here.

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¹ I. Rudnick, *J. Acoust. Soc. Am.* **25**, 1012-1013; G. C. Werth and L. P. Delsasso, *ibid.* **26**, 59-64 (1954).

² L. J. Sivian, *J. Acoust. Soc. Am.* **9**, 135-140 (1939).

³ U. Ingard and H. Ising, *J. Acoust. Soc. Am.* **42**, 6-17 (1967).

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New Laryngoscopic Technique by use of Fiber Optics

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A fiberscope with a thin flexible fiber-optics cable is inserted through the nasal passage in order to observe the laryngeal states and movements during various phonatory and articulatory conditions. Sample photographs of this procedure are presented.

A SPECIALLY DEvised, THIN, FLEXIBLE, FIBER-OPTICS CABLE WAS inserted through the nasal passage in order to observe the

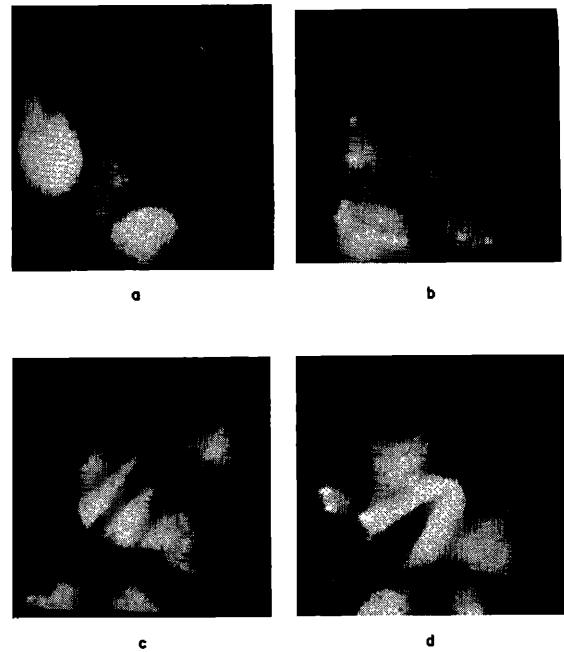


FIG. 1. Photographs of the glottis viewed through the fiberscope.

laryngeal conditions in the course of unperturbed speech utterances. Glottal conditions during various phonatory and articulatory activities have been visually inspected and photographed.

The fiberscope consists of a hard tip, a flexible optical cable, an optical connector, a light source, and a camera. The cable with its hard tip is 5.6 mm in diameter. The hard tip, 15 mm in length, houses an objective lens designed for a straightforward view with a field angle of 47° . The optical cable contains an image guide, consisting of $15\text{-}\mu$ -thick glass fibers, and a light guide, consisting of $25\text{-}\mu$ glass fibers. The optical connector contains an eye piece with a magnification factor of 11.7, and the size of the image on the photographic film is $5.6 \times 5.6 \text{ mm}^2$. The camera also houses a photosensitive element in the rear of the photographic film, so that the same fiberscope system may be used for a continuous recording of the light flux supplied transcuteaneously at the subglottal space and transmitted through the glottis. This transillumination technique is superior to previous similar methods in that the positioning of the scope can be monitored by visually inspecting the image of the glottis. A use of a motion picture in combination with a high-speed stroboscope technique is in schedule.

Before the use of the scope, the subject is anesthetized in the unilateral nasal fossa with 4% Xylocaine with Epinephrin. Unless the nasal cavity of the subject is particularly narrow, the insertion of the optical cable is carried out with ease and without causing any discomfort or disturbance to the subject in pronunciation. Light topical anesthesia may also be given to the epi- and mesopharynx.

When the tip of the scope reaches near the level of the tip of the epiglottis, the larynx is readily visible through the image finder of the camera. The scope can be kept in position with good stability by manually holding the flexible cable at the nostril. Adjustments of the positioning are also possible by external manipulation. Figure 1 illustrates sample photographs of the glottis of a normal male subject; (a) in a deep inspiration, (b) in articulation of [i] in chest register, (c) same as (b) but in head register, (d) in articulation of [ʃ]. Figure 2 illustrates, by a lateral x-ray photograph, the actual positioning of the cable.



FIG. 2. Lateral radiographic view showing positioning of the fiberoptic.

It may be noted that the glottal conditions during articulations of consonants are not visually observable in any ordinary laryngoscopic techniques. This new technique has also revealed some interesting facts about the laryngeal conditions for phonation in different registers and fundamental frequencies. Details of the experimental results will be reported in a subsequent paper.

9.7

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Effect of Masking Noise upon Syllable Duration in Oral and Whispered Reading

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In an earlier study, the whispered syllable was found to be of significantly greater duration than the oral. The present study, concerned with the effect of various levels of masking noise upon this performance, finds no effect for noise upon the magnitude of the oral-whisper difference.

IN AN EARLIER REPORT, THE AUTHOR PRESENTED THE RESULTS OF A study of the average duration of the syllable in oral and whispered reading.¹ A significant difference of 14 msec was found between the two speaking conditions (oral syllable, 243 msec; whispered syllable, 257 msec). The present report is an attempt to explain this finding.

Two models may be developed to account for the oral-whisper difference (Fig. 1). The first (A) is in terms of articulatory adjustments to sets of altered intraoral aerodynamic parameter values and suggests, for example, that individuals speak slower when whispering because they must modulate a greater volume-velocity of air through the vocal tract. Thus, according to this model, the oral-whisper difference is determined largely by a feedback system(s) whose input sensors are located within the vocal tract.

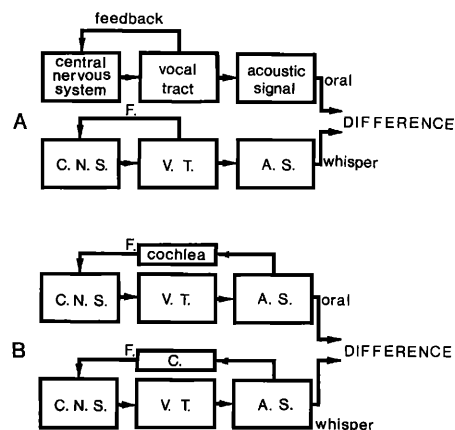


FIG. 1. Feedback models developed to account for the oral-whisper difference. Model A suggests a vocal-tract feedback system, while Model B suggests an auditory-feedback system.

The second model (B) is in terms of articulatory adjustments to reduced sidetone levels and suggests, for example, that individuals speak more slowly when whispering because they endeavor to enhance the intelligibility of a self-generated acoustic signal which is perceived as atypically weak. Thus, according to this model, the oral-whisper difference is determined largely by a feedback system whose input sensors are located within the cochlea.

It was felt that the two models could be evaluated experimentally through the simple expedient of having subjects read verbal materials in oral and whispered voice under various levels of masking noise. If the first model was correct, masking noise would not be expected to affect the oral-whisper difference.² If, on the other hand, auditory sidetone played a role in establishing the oral-whisper difference, various levels of masking noise would probably alter this difference.³

Data were obtained from 12 young adult males with normal speech and hearing. Each was seated separately in a sound-proofed room and asked to read 16 lists of five-syllable phrases. The phrases, identical to those used in the earlier experiment¹ were:

1. Push the stick forward
2. Distance from the field
3. Pressure on the stick
4. Closer to the field
5. Don't try to turn back
6. Try to stretch your glide
7. Keep a constant bank
8. Pick up too much speed
9. Dive down to the field
10. To circle the field

Phrases 1, 2, 9, and 10 occupied the same position as first and last pairs within each list; Phrases 3 through 8 were randomized separately for each list. A subject read two lists for each of three conditions of masking noise and one condition of silence for both whispered and oral voice. The order of the various auditory and speaking-voice combinations was randomized separately for each speaker.

The three levels of masking noise used were 80 dB, 65 dB, and 50 dB SPL (measured under Telephonics TDH-39 earphones, using a Brüel & Kjær, model 4152 artificial ear). The 80-dB level was selected on the basis of the results of a pilot study that showed that, at this level, the noise completely masked speakers' perceptions of their whispered sidetone. The two other noise levels were simply arbitrarily chosen at 15 and 30 dB down from this value.